Insight-V: Exploring Long-Chain Visual Reasoning with Multimodal Large Language Models

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Abstract

Multimodal large language models (MLLMs) have improved rapidly on visual question answering, yet they remain unreliable on problems that require long, branching, and self-corrective visual reasoning. A central difficulty is that long reasoning traces are both hard to curate at scale and risky to supervise directly, because errors inside the trace can corrupt the final answer model. We present Insight-V, a training framework that separates visual reasoning from answer generation. Insight-Vfirst constructs long-chain visual reasoning data with a progressive generator that emits structured intermediate steps and then filters the resulting trajectories with answer checking and path scoring. It then trains a dedicated reasoning agent on curated long traces, refines that agent with iterative direct preference optimization, and trains a separate summary agent to either use or ignore the produced reasoning when generating the final answer. Across seven visual reasoning benchmarks, Insight-Vimproves the reported average score of LLaVA-NeXT-LLaMA3-8B from 40.2 to 47.2 and of a stronger 7B base MLLM from 48.7 to 51.6. Importantly, these gains do not come at the cost of general perception: on LLaVA-NeXT, TextVQA rises from 65.2 to 70.5, DocVQA from 78.2 to 82.9, OCRBench from 553 to 663, and AI2D from 71.5 to 77.3. Ablations show that the two-agent decomposition is stronger than direct long-trace supervision, and iterative DPO yields the best reasoning performance among the tested preference-learning variants.

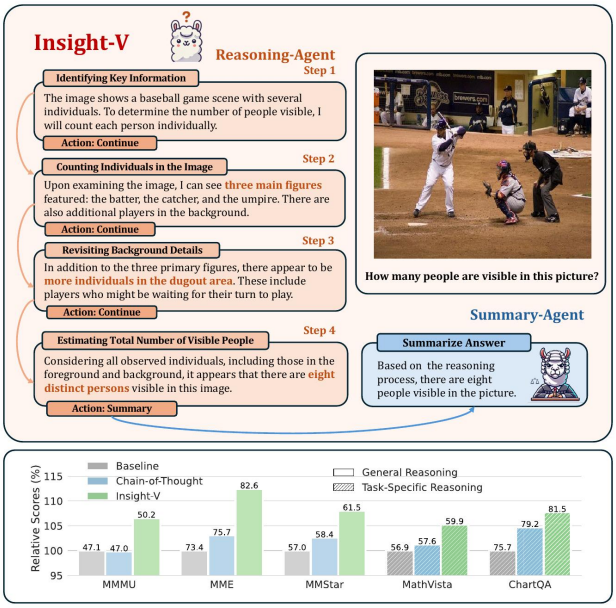

Figure 1. **Illustration and performance of Insight-V.** The framework separates long-form visual reasoning from answer summarization. On the stronger 7B base model, the largest gains appear on demanding reasoning benchmarks while preserving strong perception performance.

1. Introduction

Multimodal large language models now perform competitively on a broad range of vision-language tasks, from general instruction following to OCR-heavy reasoning and diagram understanding [3, 9–11, 15, 20]. However, their progress on *long-chain visual reasoning* is less stable. Benchmarks such as MMMU, MMMU-Pro, MathVista, ChartQA, and MMStar require a model to inspect visual evidence, test alternatives, recover from earlier mistakes, and only then summarize an answer [2, 16, 18, 28, 29]. Standard supervised fine-tuning is poorly matched to this regime: it often entangles latent reasoning with final answering and treats flawed and correct traces too similarly.

This failure mode is especially acute in multimodal settings. A long textual chain can become locally fluent while drifting away from the image. Once the model commits to an early visual mistake, the remainder of the trace frequently rationalizes that mistake instead of correcting it. Prior work on chain-of-thought and multimodal rationale generation shows the promise of explicit intermediate reasoning [24, 33], but directly optimizing long visual chains can still destabilize the underlying perception model.

Our key observation is that reasoning and answering should not be forced into the same model behavior. A reasoning model should be encouraged to explore, verify, and revise. An answering model should be trained to decide whether a given reasoning trace is trustworthy and to pro-

duce the final answer robustly even when the supplied chain is imperfect. This motivates Insight-V, a two-agent framework consisting of a *reasoning agent* and a *summary agent*. The reasoning agent learns from curated long-chain visual traces and is later refined with iterative preference optimization. The summary agent learns to condition on both the image-question pair and an optional reasoning trace enclosed by explicit `<thoughts>` tags, using that trace only when it helps.

Figure 1 summarizes the idea and its effect. On LLaVA-NeXT-LLaMA3-8B, Insight-Vimproves the reported average reasoning score from 40.2 to 47.2. On a stronger 7B base MLLM, it improves the same score from 48.7 to 51.6 while also improving or preserving performance on perception-oriented benchmarks.

Our contributions are threefold.

- We present a scalable pipeline for constructing long-chain visual reasoning data using progressive structured generation, answer filtering, and path-quality scoring.
- We introduce a two-agent MLLM training framework that disentangles reasoning from answer summarization, making the final answering stage more robust to flawed chains.
- We show that iterative preference optimization further strengthens the reasoning agent and yields consistent gains across diverse visual reasoning benchmarks without sacrificing general perception.

2. Related Work

2.1. Multimodal Large Language Models

Open MLLMs have rapidly improved through stronger vision encoders, better multimodal connectors, and more carefully curated instruction data. Representative systems include LLaVA and its successors [9–11], Qwen2-VL [20], DeepSeek-VL [15], VILA [8], Cambrian [23], InternLM-XComposer2 [3], MiniCPM-V [25, 26], Ovis [17], Idefics3 [7], Bunny [5], MM-1.5 [30], and POINTS [14]. These models demonstrate broad multimodal competence, but benchmark results in Table 1 show that even strong open models still struggle on expert reasoning datasets such as MMMU-Pro and MathVista.

2.2. Reasoning Supervision in Multimodal Models

Chain-of-thought prompting improves complex reasoning in language models [24], and multimodal variants have shown that intermediate rationales can benefit answer prediction [33]. More recent work has explored improving visual chain-of-thought behavior directly [32]. Still, the main unresolved issue is robustness: long chains can contain subtle visual mistakes, and naively supervising on those chains may amplify error propagation. Insight-Vaddresses this by separating the generation of reasoning traces from the generation of final answers and by training the final answer model to reject flawed reasoning when necessary.

2.3. Preference Optimization for Multimodal Alignment

Preference-based post-training has become a practical alternative to reinforcement learning from human feedback [1, 21]. In the multimodal setting, RLAIF-V demonstrates that open-source AI feedback can align MLLMs effectively [27]. Insight-Vadopts the simplicity of DPO but applies it iteratively to the reasoning agent, using newly sampled trajectories from the current model to create fresh preference pairs. This yields a tighter optimization loop for improving long visual reasoning instead of generic helpfulness.

3. Method

Figure 2 gives the overall design. Insight-Vhas three stages: structured long-chain data construction, reasoning-agent training, and summary-agent training.

3.1. Structured Long-Chain Data Construction

The first stage creates candidate long reasoning traces for image-question pairs. As shown in Figure 3, we use a progressive generator that produces one reasoning step at a time in a structured JSON-like format. Each step contains a step title, a content field, and an action token indicating whether the model should continue reasoning or stop and summarize. The released generation script reflects this design directly: it repeatedly prompts the model to emit a valid structured step, encourages alternative hypotheses and self-checking, and caps the trajectory length when no useful new step is produced.

Raw trajectories are noisy, so we apply two levels of assessment.

1. **Answer filtering.** Candidate responses are first checked against the ground-truth answer. The released checking script uses a stronger language model to judge semantic equivalence rather than exact string match, which is important for free-form answers.
2. **Path scoring.** Among candidates with correct final answers, an MLLM scores the reasoning quality of the path and assigns coarse quality levels. The best trajectories are retained for reasoning-agent supervision, while incorrect-answer trajectories are kept as negative context for the summary agent.

This produces two complementary datasets: a high-quality long-reasoning dataset for training a dedicated reasoning policy, and a larger summary dataset containing both useful and misleading reasoning traces. This split matches the downstream roles. The reasoning agent only observes correct, high-quality chains, so its objective stays focused on decomposition and verification rather than answer memorization. The summary agent instead learns under controlled

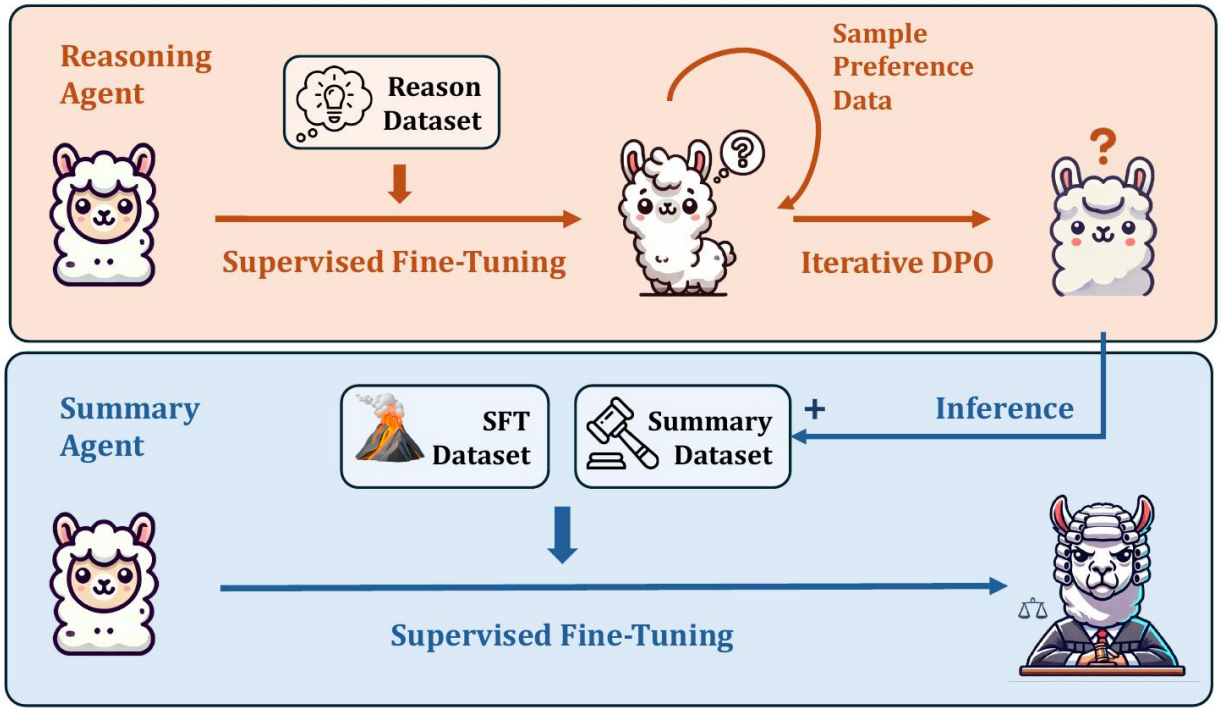

Figure 2. **Overview of Insight-V.** A reasoning agent is first trained on curated long-chain data and then improved with iterative DPO using sampled preference pairs. A separate summary agent is trained to produce the final answer from the image, the question, and an optional reasoning trace. At inference time, only the summary agent emits the answer.

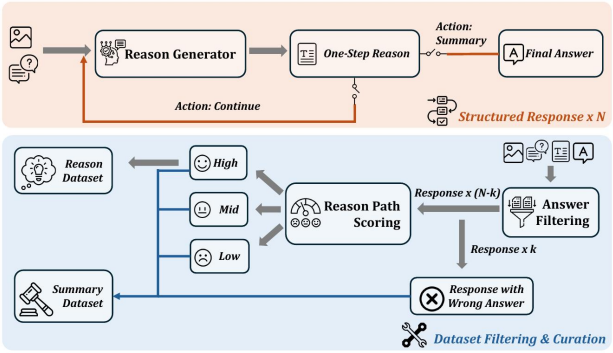

Figure 3. **Data generation pipeline of Insight-V.** Candidate structured trajectories are generated progressively, filtered by answer correctness, and then scored by path quality. High-quality correct traces are used for reasoning-agent training, while wrong-answer traces are reused to train a robust summary agent.

exposure to both trustworthy and misleading traces, which encourages selective reliance on intermediate reasoning.

3.2. Reasoning Agent

The reasoning agent is optimized to emit long-form visual reasoning enclosed by explicit `<thoughts>` tags. In the released LLaVA-NeXT implementation, the model enables a dedicated think token, uses a CLIP ViT-L/14 vision tower at 336-pixel resolution, and a two-layer MLP multimodal projector. The training scripts optimize both the multimodal connector and the language model parameters with long sequence length (12,288 tokens), which is necessary for multi-step visual chains.

The first phase is supervised fine-tuning on the curated reasoning set. Let x denote an image-question pair and r^+ a curated reasoning trace. The reasoning agent with parameters θ is trained with standard teacher forcing:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,r^+)} \log p_\theta(r^+ \mid x). \tag{1}$$

This stage teaches explicit decomposition and verification behavior, but it does not explicitly distinguish between merely plausible and genuinely strong reasoning paths.

3.3. Iterative Preference Optimization

To sharpen the reasoning policy, Insight-Vperforms iterative DPO on sampled pairs of trajectories [21]. At each round, the current reasoning agent generates new candidate chains. After filtering and scoring, we construct preference pairs (r^+, r^-) where r^+ is preferred over r^- for the same input x.

We then optimize

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,r^+,r^-)} \log \sigma\Big(\beta \log \frac{p_\theta(r^+ \mid x)}{p_\theta(r^- \mid x)} - \beta \log \frac{p_{\text{ref}}(r^+ \mid x)}{p_{\text{ref}}(r^- \mid x)}\Big) \tag{2}$$

where p_{ref} is the frozen reference model and β is the DPO temperature. Unlike a single-pass preference stage, iterative DPO refreshes the preference data using the improved current model, which better targets the model's remaining reasoning errors.

3.4. Summary Agent

The summary agent receives the image, the original question, and optionally a reasoning trace surrounded by `<thoughts>` tags. Its job is not to imitate long reasoning. Instead, it must decide whether the provided reasoning is useful and then produce the final answer directly. During training, the summary dataset mixes three cases: correct high-quality traces, flawed traces, and general visual instruction data without reasoning. This mixture regularizes the summary model so that it does not become over-dependent on any chain.

At inference time, the system runs in two stages. The reasoning agent first emits a long trace. The summary agent then conditions on the image, the question, and that trace, and produces the final answer. This decomposition is important: the reasoning model is free to explore, while the summary model is optimized for answer robustness.

4. Experiments

4.1. Experimental Setup

We evaluate Insight-Von two backbones. The first is LLaVA-NeXT-LLaMA3-8B [10]. The second is a stronger 7B base MLLM, denoted as *Our Base Model* in the released tables. For the LLaVA-NeXT implementation, the released scripts indicate a CLIP ViT-L/14 336px vision tower, a two-layer MLP projector, bf16 training, and joint tuning of the connector and language model. Reasoning-agent supervised fine-tuning and DPO both use two epochs with a learning rate of 5×10^{-6}, while summary-agent fine-tuning uses one epoch with a learning rate of 10^{-5}.

We report results on seven reasoning benchmarks: MMMU [28], MMMU-Pro [29], MMBench [12], MME [4], ChartQA [18], MMStar [2], and MathVista [16]. We additionally evaluate four perception-oriented benchmarks: TextVQA [22], DocVQA [19], OCRBench [13], and AI2D [6]. Following the project artifacts, the evaluations are run with `lmms-eval`, and MathVista is judged with GPT-4-0613.

4.2. Main Results on Visual Reasoning

Table 1 reports the main reasoning results. Insight-Vimproves the reported average score of LLaVA-NeXT-LLaMA3-8B from 40.2 to 47.2, a gain of 7.0 points. The gains are broad rather than isolated: MMMU rises from 36.9 to 42.0, MMMU-Pro from 13.2 to 21.0, MMBench from 72.3 to 81.7, ChartQA from 69.4 to 77.4, MMStar from 43.1 to 57.4, and MathVista from 45.9 to 49.8. These are substantial improvements for a backbone that is otherwise well behind the strongest open baselines on expert reasoning tasks.

The stronger 7B base model also benefits, although the gains are naturally smaller because the starting point is stronger. Its reported average increases from 48.7 to 51.6. We observe consistent gains on MMMU (47.1 $\rightarrow$ 50.2), MMMU-Pro (22.6 $\rightarrow$ 24.9), ChartQA (75.7 $\rightarrow$ 81.5), MMStar (57.0 $\rightarrow$ 61.5), and MathVista (56.9 $\rightarrow$ 59.9), with MMBench reaching the best value in the table for this model family at 82.3.

The MME results are mixed for the LLaVA-NeXT backbone and positive for the stronger base model. On LLaVA-NeXT, the second MME score rises markedly from 346.0 to 485.4 while the first score declines from 1611.1 to 1583.9. On the stronger 7B base, both MME scores improve to 1685.1 and 627.0. This pattern is consistent with the broader claim of the paper: Insight-Vprimarily targets long-form reasoning and does not guarantee uniform improvement on every aggregate diagnostic.

4.3. Results on Perception-Oriented Benchmarks

Table 2 evaluates whether stronger reasoning hurts general perception. For LLaVA-NeXT, the answer is clearly no: TextVQA improves from 65.2 to 70.5, DocVQA from 78.2 to 82.9, OCRBench from 553 to 663, and AI2D from 71.5 to 77.3. The summary-agent training appears important here because it exposes the model to both helpful and unhelpful reasoning traces instead of forcing reliance on every chain.

For the stronger 7B base model, perception remains strong after applying Insight-V, with the best DocVQA result (91.5) and near-best OCRBench and AI2D values. One nuance is that the intermediate multi-agent stage slightly outperforms iterative DPO on TextVQA, OCRBench, and AI2D for this backbone. This reinforces that preference optimization mainly improves reasoning quality; its effect on general perception is smaller and not always monotonic.

5. Analysis

5.1. Why the Two-Agent Design Helps

Table 3 compares alternative designs on four representative reasoning benchmarks. Direct SFT on long traces improves over the baseline (59.2 $\rightarrow$ 60.6), and multi-turn supervision does slightly better (61.0), but the full two-agent design reaches 62.1. The summary-only variant underperforms at 59.8, indicating that the gains do not come merely from training a better answer head. The main benefit comes from

Table 1. **Results on Visual Reasoning Tasks.** We conduct evaluation experiments across 7 benchmarks, covering both general reasoning and task-specific reasoning assessments. Insight-V exhibits notable effectiveness and generalizability when applied to LLaVA-NeXT and our baseline model, surpassing other state-of-the-art MLLMs by a large margin.

Model	Size	MMMU	MMMU-Pro	MMBench	MME	ChartQA	MMStar	MathVista	Average
DeepSeek-VL [15]	7B	35.4	-	73.5	-/-	59.1	37.1	36.1	-
VILA-1.5 [8]	8B	38.6	-	75.3	1634.9/-	-	39.7	-	-
Cambrian-1 [23]	8B	42.7	-	75.9	1547.1/-	73.3	-	49.0	-
InternLM-XComposer2 [3]	7B	41.1	-	77.6	2220.4	71.8	56.2	59.5	-
POINTS [14]	7B	51.4	-	78.0	2184.1	-	60.9	63.0	-
IXC-2.5 [31]	7B	42.9	-	79.4	2233.1	82.2	59.9	**63.7**	-
Bunny-LLaMA3 [5]	8B	43.4	-	77.2	1588.9/321.1	-	-	34.4	-
MM-1.5 [30]	7B	41.8	-	-	1514.9/346.4	78.6	-	47.6	-
MiniCPM-LLaMA3-V 2.5 [25]	8B	45.8	19.6	77.2	2024.6	-	51.8	54.3	-
MiniCPM-V-2.6 [26]	7B	49.8	**27.2**	78.0	2268.7	-	57.5	60.6	-
Qwen2-VL [20]	7B	**53.7**	-	81.0	-	**83.0**	60.7	61.4	-
Idefics3-LLaMA3 [7]	8B	46.6	22.9	77.5	1937.4	74.8	55.9	58.4	48.1
Ovis1.5-LLaMA3 [17]	8B	48.3	23.6	76.6	1948.5	76.4	57.3	63.0	49.4
LLaVA-NeXT-LLaMA3 [10]	8B	36.9	13.2	72.3	1611.1/346.0	69.4	43.1	45.9	40.2
+ Multi-Agent	8B	40.8	17.8	77.6	1603.7/469.3	74.6	52.6	47.4	44.5
+ Iterative DPO (**Insight-V-LLaVA**)	8B	42.0	21.0	81.7	1583.9/485.4	77.4	57.4	49.8	**47.2** (+7.0)
Our Base Model	7B	47.1	22.6	81.3	1573.7/482.5	75.7	57.0	56.9	48.7
+ Multi-Agent	7B	49.7	23.8	82.2	1662.2/**629.3**	81.2	58.6	58.7	50.7
+ Iterative DPO (**Insight-V**)	7B	50.2	24.9	**82.3**	**1685.1**/627.0	81.5	**61.5**	59.9	**51.6** (+2.9)

Table 2. **Results on other multimodal benchmarks.** Insight-V enhances reasoning capabilities without compromising general visual perception and even achieves improvements on benchmarks requiring perception ability more.

Model	TextVQA	DocVQA	OCRBench	AI2D
LLaVA-NeXT-LLaMA3	65.2	78.2	553	71.5
+ Multi-Agent	68.9	81.8	631	75.7
+ Iterative DPO (**Insight-V-LLaVA**)	**70.5**	**82.9**	**663**	**77.3**
Our Base Model	75.4	90.2	713	79.7
+ Multi-Agent	**77.0**	91.4	**738**	**80.1**
+ Iterative DPO (**Insight-V**)	76.8	**91.5**	735	79.8

Table 3. **Ablations on the Insight-V Design Choice.** The multi-agent design outperforms other configurations, highlighting the critical role of reasoning and summarization decomposition.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Baseline	47.1	75.7	56.9	57.0	59.2
Vanilla - Direct SFT	47.0	79.2	57.6	58.4	60.6
Multi-Turn Supervised	48.1	79.6	57.9	58.2	61.0
Summary Agent Only	47.5	76.3	57.3	57.9	59.8
Multi-Agent	49.7	81.2	58.7	58.6	62.1

Table 4. **Ablations on the DPO training strategy.** Iterative DPO progressively enhances the model's reasoning capabilities, leading to improved performance.

Model	MMMU	ChartQA	MathVista	MMStar	Avg
Insight-V (Multi-Agent)	49.7	81.2	58.7	58.6	62.1
+ RLAIF	49.5	81.4	59.1	59.2	62.3
+ DPO	50.8	80.8	59.3	59.9	62.7
+ Iterative DPO	50.2	81.5	59.9	61.5	63.3

learning a dedicated reasoning policy and then letting a separate answer model decide how much to trust it.

5.2. Why Iterative DPO Helps

Table 4 shows that iterative DPO yields the best average result among the tested alignment strategies. Starting from the multi-agent model at 62.1 average, RLAIF raises performance to 62.3 and single-pass DPO to 62.7, while iterative DPO reaches 63.3. The largest gain appears on MMStar (58.6 $\rightarrow$ 61.5), suggesting that repeated preference refinement is particularly helpful when tasks require visually grounded elimination of distractors rather than shallow pattern matching.

Figure 4 normalizes five representative benchmarks to the baseline model and makes the same pattern visible from a different angle: direct chain-of-thought supervision helps, but the full two-agent pipeline produces larger gains once the task requires explicit verification across several intermediate decisions.

5.3. Scaling with More Reasoning Data

Figure 5 studies the amount of curated reasoning data. Performance is reported relative to the baseline model. At 50K examples, several benchmarks remain close to or slightly below the baseline, which shows that noisy or insufficient reasoning data is not enough. After 100K examples, all reported tasks move above the baseline, and scaling to 150K and 200K continues to improve performance. The strongest growth appears on MME and ChartQA, while MMMU, MMMU-Pro, MathVista, and MMStar improve more steadily. This trend supports the paper's central claim that long-chain reasoning needs both quality control and scale.

5.4. Qualitative Behavior

Figure 6 illustrates a typical failure case for direct long-trace supervision. The monolithic model starts correctly but then

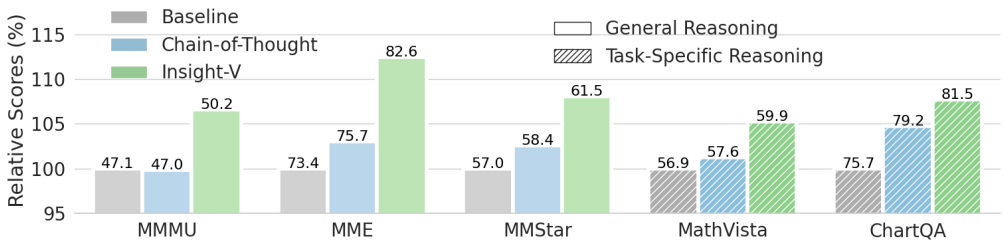

Figure 4. **Relative comparison on representative benchmarks.** Insight-Vconsistently exceeds both the baseline and direct chain-of-thought supervision, with the clearest gains on MME, MMStar, and ChartQA.

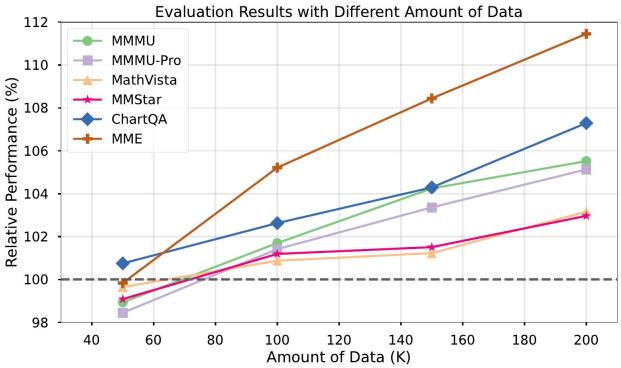

Figure 5. **Ablation on the amount of training data.** Relative performance is normalized so that 100 corresponds to the baseline model. Gains become consistently positive once the curated reasoning set reaches 100K examples, and larger datasets continue to help.

compounds an intermediate arithmetic or attribution mistake and commits to the wrong final option. In contrast, the reasoning agent in Insight-Vproceeds through explicit verification steps, and the summary agent produces the final answer only after consuming the validated chain. The example is consistent with the quantitative ablations: decoupling reasoning from answering reduces error propagation.

6. Conclusion

We presented Insight-V, a framework for long-chain visual reasoning in multimodal large language models. The method combines progressive structured data generation, quality-aware filtering, a dedicated reasoning agent, a separate summary agent, and iterative preference optimization. Across two backbones and a diverse evaluation suite, Insight-Vdelivers strong improvements on difficult visual reasoning benchmarks while preserving, and often improving, perception performance. The overall evidence suggests that separating reasoning from final answer generation is a practical and effective design principle for robust multimodal reasoning.

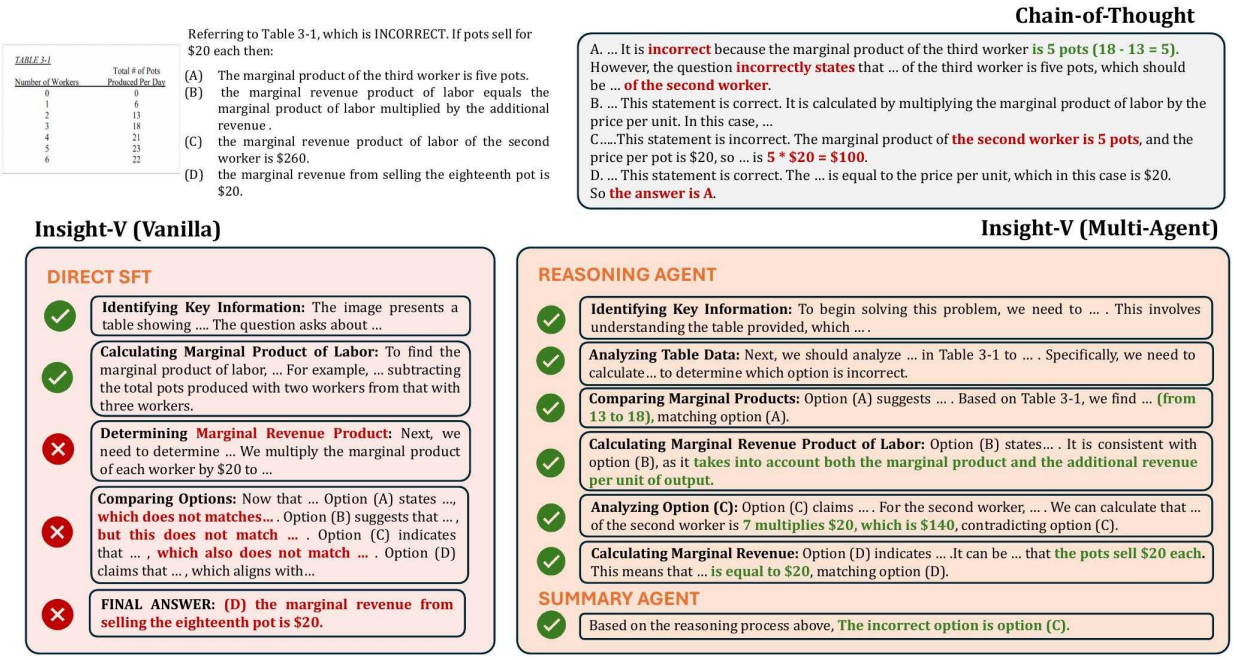

Figure 6. **Qualitative results of Insight-V.** A monolithic direct-SFT model begins with partially useful reasoning but drifts into an incorrect answer. The two-agent pipeline maintains explicit verification steps and allows the summary agent to output the answer supported by the validated chain.

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